

# Yuzong Chen

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## EDUCATION

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- **Cornell University** Aug. 2022 – Aug. 2026 (Expected)  
Ph.D. in Electrical and Computer Engineering  
*Advisor: Prof. Mohamed S. Abdelfattah*
- **Nanyang Technological University, Singapore** Aug. 2015 – Jun. 2019  
B.Eng. in Electrical & Electronic Engineering  
GPA: 4.74 / 5.00, Honours (Highest Distinction)

## RESEARCH INTERESTS

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**Efficient Hardware for Machine Learning:** I am interested in developing efficient hardware for accelerating machine learning. This includes inference algorithms optimization and accelerator co-design.

## RESEARCH EXPERIENCE

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- **Cornell University** Aug. 2022 – Present  
*Graduate Research Assistant* *Advisor: Prof. Mohamed S. Abdelfattah*  
Work on algorithm-hardware co-design for machine learning acceleration, with a special focus on quantization numerics, reconfigurable architectures, and processing in-memory.
- **Stanford University** Jan. 2026 – Apr. 2026  
*Visiting Scholar* *Advisor: Prof. Thierry Tamba*  
Work on large language model acceleration via compute-optimal test-time scaling.
- **National University of Singapore** Sept. 2021 – Jul. 2022  
*Research Engineer* *Advisor: Prof. Heng Chun-Huat*  
Work on a joint project with NXP Semiconductors to design an RF switched-capacitor power amplifier for high-speed communication. Help tape-out the chip in 22nm FDSOI technology.
- **Nanyang Technological University, Singapore** Feb. 2020 – Aug. 2021  
*Project Officer* *Advisor: Prof. Tony Tae-Hyoung Kim*  
Conduct and lead projects about processing in-memory circuit design based on static random access memory (SRAM) and resistive random access memory (ReRAM). Tape-out several chips in 65nm technology.

## INDUSTRY EXPERIENCE

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- **Tesla Inc.** May 2026 – Present  
*Senior Autopilot Engineer* *Palo Alto, CA*  
Co-design the next generation of AI chips for Tesla Optimus and Full Self-Driving.
- **Samsung Semiconductor** Jun. 2025 – Aug. 2025  
*Architecture Research Intern* *San Jose, CA*  
Propose novel mixed-precision LLM quantization algorithms by jointly optimizing linear and attention layers.
- **Qualcomm AI Research** Sept. 2024 – Dec. 2024  
*Research Intern* *San Diego, CA*  
Implement performance modelling framework for LLM inference on heterogeneous NPU-PIM systems.

## PUBLICATIONS

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- **Conference Proceedings** († Equal Contribution)
- [1] **Yuzong Chen**<sup>†</sup>, Xilai Dai<sup>†</sup>, Jake Hyun<sup>†</sup>, Chi-Chih Chang, Wonsuk Jang, Yuheng Wu, Thierry Tamba, Jae-sun Seo, Mohamed S. Abdelfattah, “[RaZeR: Pushing the Limits of NVFP4 Quantization with Redundant Zero Remapping](#)”, *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
  - [2] **Yuzong Chen**, Chao Fang, Xilai Dai, Yuheng Wu, Thierry Tamba, Marian Verhelst, Mohamed S. Abdelfattah, “[Unlocking Efficient Processing-In-Memory for Edge LLM Inference with Hybrid Numerical Formats](#)”, *IEEE/ACM International Symposium on Computer Architecture (ISCA)*, 2026.
  - [3] **Yuzong Chen**, Ahmed AbouElhamayed, Xilai Dai, Yang Wang, Marta Andronic, George A. Constantinides,

- Mohamed S. Abdelfattah, “[BitMoD: Bit-serial Mixture-of-Datatype LLM Acceleration](#)”, *IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, 2025.
- [4] **Yuzong Chen**, Jian Meng, Jae-sun Seo, Mohamed S. Abdelfattah, “[BBS: Bi-directional Bit-level Sparsity for Deep Learning Acceleration](#),” *IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 2024.
  - [5] Xilai Dai, **Yuzong Chen**, Mohamed S. Abdelfattah, “[Kratos: An FPGA Benchmark for Unrolled Deep Neural Networks with Fine-Grained Sparsity and Mixed Precision](#)”, *IEEE International Conference on Field-Programmable Logic and Applications (FPL)*, 2024.
  - [6] Jordan Dotzel, **Yuzong Chen**, Bahaa Kotb, Sushma Prasad, Gang Wu, Sheng Li, Mohamed S. Abdelfattah, Zhiru Zhang, “[Learning from Students: Applying t-Distributions to Explore Accurate and Efficient Formats for LLMs](#)”, *International Conference on Machine Learning (ICML)*, 2024.
  - [7] **Yuzong Chen**, Jordan Dotzel, Mohamed S. Abdelfattah, “[M4BRAM: Mixed-Precision Matrix-Matrix Multiplication in FPGA Block RAMs](#)”, *IEEE International Conference on Field Programmable Technology (FPT)*, 2023.
  - [8] **Yuzong Chen**, Mohamed S. Abdelfattah, “[BRAMAC: Compute-in-BRAM Architectures for Multiply-Accumulate on FPGAs](#)”, *IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM)*, 2023.
  - [9] **Yuzong Chen**, Junjie Mu, Hyunjoon Kim, Lu Lu, Tony Tae-Hyoung Kim, “[A Reconfigurable 8T SRAM Macro for Bit-Parallel Searching and Computing In-Memory](#)”, *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2022.
  - [10] Yuncheng Lu, Zehao Li, **Yuzong Chen**, Tony Tae-Hyoung Kim, “[A 181 \$\mu\$ W Real-Time 3-D Hand Gesture Recognition System based on Bi-directional Convolution and Memoryless Clustering](#)”, *IEEE Custom Integrated Circuits Conference (CICC)*, 2022.
  - [11] **Yuzong Chen**, Lu Lu, Yuncheng Lu, Tony Tae-Hyoung Kim, “[A Multi-Functional 4T2R ReRAM Macro Enabling 2-Dimensional Access and Computing In-Memory](#)”, *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2021.
  - [12] Lu Lu, **Yuzong Chen**, Tony Tae-Hyoung Kim, “[A Configurable Randomness Enhanced RRAM PUF with Biased Current Sensing Scheme](#)”, *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2021.
  - [13] Vishal Sharma, Ju Eon Kim, Yong-Jun Jo, **Yuzong Chen**, Tony Tae-Hyoung Kim, “[AND8T SRAM Macro with Improved Linearity for Multi-bit In-Memory Computing](#)”, *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2021.
  - [14] **Yuzong Chen**, Lu Lu, Bongjin Kim, and Tony Tae-Hyoung Kim, “[Reconfigurable 2T2R ReRAM with Split Word-lines for TCAM Operation and In-Memory Computing](#)”, *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2020.

#### • Journal Articles

- [1] **Yuzong Chen**, Chi-Chih Chang, Xilai Dai, Ahmed AbouElhamayed, Marta Andronic, George A. Constantinides, Mohamed S. Abdelfattah “[Bit-serial Acceleration of LLM Inference with Mixture-of-Datatype Quantization](#)”, *IEEE Transactions on Computers (TC)*, 2025. [Selected as Featured Paper in the February 2026 issue of IEEE TC]
- [2] **Yuzong Chen**, Junjie Mu, Hyunjoon Kim, Lu Lu, Tony Tae-Hyoung Kim, “[BP-SCIM: A Reconfigurable 8T SRAM Macro for Bit-Parallel Searching and Computing In-Memory](#)”, *IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)*, 2023.
- [3] Donghyuk Kim, Chengshuo Yu, Shanshan Xie, **Yuzong Chen**, Joo-Young Kim, Bongjin Kim, Jaydeep Kulkarni, Tony Tae-Hyoung Kim, “[An Overview of Processing-in-Memory Circuits for Artificial Intelligence and Machine Learning](#)”, *IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS)*, 2022. [Featured as one of the top-3 popular articles in JETCAS 2025]
- [4] **Yuzong Chen**, Lu Lu, Bongjin Kim, Tony Tae-Hyoung Kim, “[A Reconfigurable 4T2R ReRAM Computing In-Memory Macro for Efficient Edge Applications](#)”, *IEEE Open Journal of Circuits and Systems (OJCAS)*, 2021.
- [5] **Yuzong Chen**, Lu Lu, Bongjin Kim, Tony Tae-Hyoung Kim, “[Reconfigurable 2T2R ReRAM Architecture for Versatile Data Storage and Computing In-Memory](#)”, *IEEE Transactions on Very Large Scale Integration Systems (TVLSI)*, 2020.

#### TEACHING EXPERIENCE

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|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Modern Computer Systems and Architecture (ECE5755 / CS5754, Cornell Tech)</li> </ul> | Fall 2025<br><i>Instructor: Prof. Udit Gupta</i> |
| <ul style="list-style-type: none"> <li>• Applied Machine Learning (CS5785, Cornell Tech)</li> </ul>                           | Fall 2023<br><i>Instructor: Prof. Kyra Gan</i>   |

## HONOURS AND AWARDS

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- MLCommons ML and Systems Rising Star, 2026
- IEEE TC Featured Paper of the Month, February 2026
- Student Travel Grant: HPCA 2025
- IEEE JETCAS Top Three Popular Articles, 2025
- Qualcomm Innovation Fellowship (QIF), Finalist, 2024
- Cornell Graduate Fellowship, 2022
- NTU Singapore Undergraduate Dean's List (top 5% of the cohort), 2015 – 2017
- Singapore Science and Engineering Undergraduate Scholarship, 2015 – 2019

## INVITED TALKS

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- **Efficient Algorithm-Hardware Co-Design Methodology for Quantized LLM Acceleration**
  - Stanford Portal Seminar. Host: Prof. [Thierry Tamba](#) (Jan. 2026)
  - UCF Computer Architecture Seminar Series. Host: Prof. [Di Wu](#) (Mar. 2026)
  - Workato AI Journal Club. Host: Prof. [Zhaozhuo Xu](#) (May. 2026)
- **Leveraging Bit-serial Computation for Deep Learning Acceleration**
  - Samsung AI Research. Host: Dr. [Stylianos I. Venieris](#) (Jul. 2024)
  - MICRO'24 (Nov. 2024)
  - Southeast University. Host: Prof. [He Li](#) (Dec. 2024)
- **BitMoD: Bit-serial Mixture-of-Datatype LLM Acceleration**
  - Qualcomm AI Research. Host: Dr. [Paul Whatmough](#) (Nov. 2024)
  - HPCA'25 (Mar. 2025)
  - Computer Architecture Day @ Columbia University. Host: Prof. [Martha Kim](#) (May 2025)
- **Efficient Computing In-memory Architectures for FPGA-based Deep Learning Acceleration**
  - FCCM'23 (May 2023)
  - FPT'23 (Dec. 2023)
  - Centre for Spatial Computational Learning. Host: Prof. [George A. Constantinides](#) (Mar. 2024)

## EDITORIAL SERVICE

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- **Journal Reviewer**
  - IEEE Transactions on Computers (TC)
  - IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)
  - IEEE Transactions on Very Large Scale Integration Systems (TVLSI)
  - IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS)
- **Conference Secondary Reviewer**
  - International Symposium on Field-Programmable Gate Arrays (FPGA), 2024
  - Design Automation Conference (DAC), 2023
  - Conference on Machine Learning and Systems (MLSys), 2025
  - International Symposium on High-Performance Computer Architecture (HPCA), 2027
- **Artifact Evaluation Committee**
  - International Symposium on High-Performance Computer Architecture (HPCA), 2025